WASP-3b

R-1026

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-3_161030` · created 2026-07-30T05:26:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457691.6209 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.033
Transit depth (Rp/Rs)^2	4.9e-06 +/- 0.0015 [%]
Orbital Inclination (inc)	86.94 +/- 2.73
Airmass coefficient 1 (a1)	7.54 +/- 2.75
Airmass coefficient 2 (a2)	-1.543 +/- 0.066
Scatter in the residuals of the lightcurve fit is	23.86 %
Best Comparison Star	#1 - [200, 30]
Optimal Aperture	5.68
Optimal Annulus	11.83
Transit Duration (day)	0.0907 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.033 vs 0.10538 (deviation 0.52σ)
Duration fit vs published	0.0907 d vs 0.1167 d (deviation 0.84σ)
Mid-time O–C	-0.8 min
Depth SNR	0.0
Residual scatter	23.86 %

Light curve

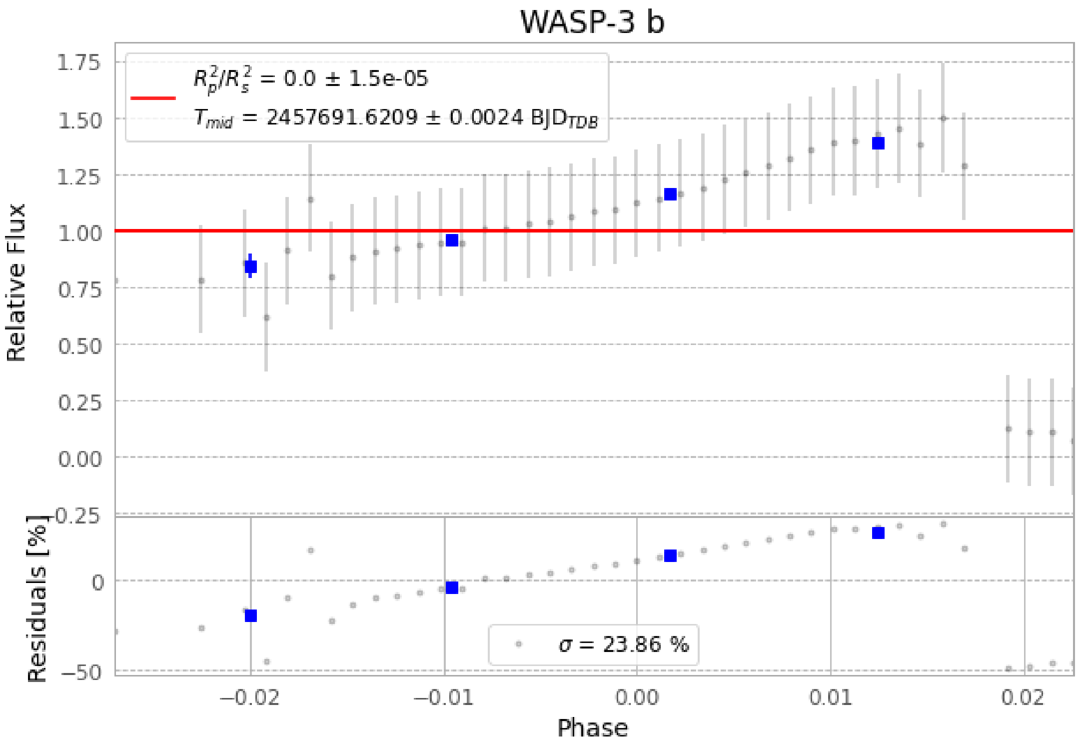

Figure 1 — FinalLightCurve_WASP-3 b_2016-10-29.png (SHA-256 06f7b7a06dc550f7…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [597, 247], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 f782fb7531444842…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f8439d0`

Data provenance

86 FITS frames (56 MB), `WASP-3161030014215.FITS` → `WASP-3161030060312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-161030.log (da0cc41e4394…), FinalLightCurve_WASP-3 b_2016-10-29.png (06f7b7a06dc5…), FinalLightCurve_WASP-3 b_2016-10-29.csv (3eda8022d827…)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)